Solutions – Practice Problem Set 2

ECON 101 – Macroeconomics

Fall 2025

1. A study that makes coffee more attractive increases consumers' willingness to pay at each price.
   Graph: Demand shifts right from D_0 to D_1; supply S is unchanged.
   New equilibrium. The intersection of D_1 and S implies a higher equilibrium price $P_1 > P_0$ and higher equilibrium quantity $Q_1 > Q_0$.
   Explanation: At the original price P_0, the rightward shift creates excess demand, bidding up price. The price rises until the new intersection clears the market.

2. A cost-reducing technology raises firms' willingness to supply at each price.
   Graph: Supply shifts right from S_0 to S_1; demand D is unchanged.
   New equilibrium: The intersection of S_1 and D implies a lower equilibrium price $P_1 < P_0$ and higher equilibrium quantity $Q_1 > Q_0$.
   Explanation: At the original price P_0, the rightward shift creates excess supply, pushing price down. The price falls until the new intersection clears the market.

3. $$Q_d = 100 - 2P, \qquad Q_s = 20 + 3P.$$

   Equilibrium. Set $Q_d = Q_s$:

   $$100 - 2P = 20 + 3P \;\Rightarrow\; 80 = 5P \;\Rightarrow\; P^* = 16.$$

   $$Q^* = 100 - 2(16) = 100 - 32 = 68 \quad \text{(checks: } 20 + 3 \cdot 16 = 20 + 48 = 68\text{)}.$$

   At $P = 10$.
   $$Q_d = 100 - 2(10) = 80, \qquad Q_s = 20 + 3(10) = 50.$$

   Result. Shortage of $80 - 50 = 30$ units.

4. $$Q_d = 100 - 2(20) = 100 - 40 = 60, \qquad Q_s = 20 + 3(20) = 20 + 60 = 80.$$

   Surplus. $80 - 60 = 20$ units.
   Binding? Yes. The floor 20 exceeds the equilibrium price 16, so it prevents the market from reaching P^* and creates excess supply.

5.

$$Q_d = 80 - 2P, \qquad Q_s = 10 + 2P.$$

Equilibrium without ceiling. Set $Q_d = Q_s$:

$$80 - 2P = 10 + 2P \;\Rightarrow\; 70 = 4P \;\Rightarrow\; P^* = 17.5.$$

$$Q^* = 80 - 2(17.5) = 80 - 35 = 45 \quad \text{(checks: } 10 + 2(17.5) = 10 + 35 = 45\text{)}.$$

With ceiling $P^c = 15$:

$$Q_d = 80 - 2(15) = 80 - 30 = 50, \qquad Q_s = 10 + 2(15) = 10 + 30 = 40.$$

Shortage. $50 - 40 = 10$ units.
Possible unintended consequence. Reduced maintenance and quality of rental units as landlords face lower effective prices and weaker incentives to invest.

6. **Expenditure Approach**

Data (billions): $C = 1400$, $I = 500$, $G = 600$, $X = 400$, $M = 450$.

(a) GDP using expenditure approach

$$\text{GDP} = C+I+G+(X-M) = 1400+500+600+(400-450) = 1400+500+600-50 = \boxed{2450}$$

(billions of dollars)

(b) NNP and GNP given depreciation $= 200$*, net factor payments from abroad* $= NFP = -50$

$$\text{GNP} = \text{GDP} + \text{NFP} = 2450 + (-50) = \boxed{2400}$$

$$\text{NNP} = \text{GNP} - \text{Depreciation} = 2400 - 200 = \boxed{2200}$$

(all in billions of dollars)

7. **Income Approach**

Data (billions): Wages & Salaries = 2200, Rental Income = 300, Corporate Profits = 400, Net Interest = 200, Indirect Taxes less Subsidies = 250, Depreciation = 150.

(a) GDP via income approach

$$\text{GDP} = \underbrace{(2200 + 300 + 400 + 200)}_{\text{factor incomes}} + \underbrace{250}_{\text{indirect taxes}-\text{subsidies}} + \underbrace{150}_{\text{depreciation}} = 3500$$

$$\boxed{\text{GDP} = 3500 \text{ (billions)}}$$

(b) Compare to expenditure GDP of \$3,350b

$$\text{Statistical discrepancy} = 3500 - 3350 = \boxed{150 \text{ (billions)}}$$

Interpretation: measurement and timing differences across sources leave a positive discrepancy of \$150b.

8. **Real vs Nominal GDP**

Two goods: bread and smartphones.

Year	Bread (Qty)	Bread Price	Smartphones (Qty)	Phone Price
2023	100	\$2.00	50	\$400
2024	110	\$2.20	60	\$380

(a) Nominal GDP

$$\text{Nominal GDP}_{2023} = 100 \times 2 + 50 \times 400 = 200 + 20000 = \boxed{20200}$$

$$\text{Nominal GDP}_{2024} = 110 \times 2.20 + 60 \times 380 = 242 + 22800 = \boxed{23042}$$

(b) Real GDP (base year = 2023)

$$\text{Real GDP}_{2023}^{(2023\text{ prices})} = \boxed{20200}$$

$$\text{Real GDP}_{2024}^{(2023\text{ prices})} = 110 \times 2.00 + 60 \times 400 = 220 + 24000 = \boxed{24220}$$

(c) GDP deflator, 2024

$$\text{Deflator}_{2024} = \frac{\text{Nominal}_{2024}}{\text{Real}_{2024}} \times 100 = \frac{23042}{24220} \times 100 \approx \boxed{95.14}$$

(2023 base = 100 implies an overall price decline driven by cheaper smartphones.)

9. **CPI Calculation**

Basket: 10 bread, 5 movie tickets.

(a) CPI in 2023 and 2024

$$\text{Cost}_{2023} = 10 \times 2 + 5 \times 10 = 20 + 50 = 70 \quad \Rightarrow \quad \text{CPI}_{2023} = \boxed{100}$$

$$\text{Cost}_{2024} = 10 \times 2.20 + 5 \times 12 = 22 + 60 = 82 \quad \Rightarrow \quad \text{CPI}_{2024} = \frac{82}{70} \times 100 \approx \boxed{117.14}$$

(b) Inflation rate (CPI), 2023 to 2024

$$\pi = \frac{\text{CPI}_{2024} - \text{CPI}_{2023}}{\text{CPI}_{2023}} \times 100 = \frac{117.14 - 100}{100} \times 100 \approx \boxed{17.14\%}$$

(c) Compare CPI inflation with GDP deflator from Q3

$$\text{GDP deflator change} \approx 95.14 - 100 = -4.86 \Rightarrow \text{deflation of } \boxed{4.86\%}$$

Differences arise because:

- **Coverage**: CPI tracks consumer purchases, includes imports; GDP deflator covers domestically produced final goods and services, excludes imports and includes exports.
- **Weights**: CPI fixes a basket; GDP deflator uses current production weights.
- **Relative price movements**: Large smartphone price declines lower the deflator more, while the CPI basket assigns different weights and may reflect higher movie-ticket inflation.

10. **Nominal vs Real Wages**

CPI: 100 $\to$ 110. Nominal wage: \$20 (2023) $\to$ \$21 (2024).

(a) Real wage in 2024 (base 2023=100)

$$w_{2024}^{\text{real}} = \frac{w_{2024}^{\text{nom}}}{\text{CPI}_{2024}/100} = \frac{21}{1.10} \approx \boxed{\$19.09}$$

(with base-year purchasing power)

(b) Did purchasing power rise? In 2023, real wage was \$20 (base). In 2024, it is about \$19.09, so purchasing power *fell*. The nominal raise did not keep up with inflation.